

**Post-MDT Oncology Workflow
Bottlenecks: A Design-Science Study
of an NHS Integrated Data Repository
and Decision-Support Dashboard to
Reduce Delays from MDT Outcome to
Treatment Start.**

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Chapter 1

Introduction

Delivering timely cancer care remains a critical priority within the NHS, where delays at any stage of the diagnostic and treatment pathway can negatively affect patient outcomes, experience, and service performance. Recent policy reforms have consolidated the NHS Cancer Waiting Time (CWT) framework into three outcome-focused standards: the 28-day Faster Diagnosis Standard (FDS), the 31-day decision-to-treat to treatment (DTT) standard, and the 62-day referral-to-treatment (RTT) standard—implemented nationally from October 2023 (Johnson, 2024; NHS Digital, 2026). These standards place increased emphasis on operational viability, rapid decision-making, and efficient progression of patients through the post-MDT (multidisciplinary team) stages of the oncology pathway. However, many Trusts continue to face systemic challenges in generating timely, accurate, and actionable insights from fragmented data sources.

Within Portsmouth Hospitals University NHS Trust, as in many NHS institutions, key operational data concerning MDT outcomes, clinic scheduling, booking practices, and treatment initiation remain dispersed across siloed systems. Although these systems individually support MDT coordination, outpatient clinic management, and treatment delivery, they lack a consolidated view of pathway performance. This fragmentation makes it difficult for operational managers to detect bottlenecks, identify emerging breach risks, or allocate clinical capacity efficiently. As a result, delays between **MDT outcomes**, **first oncology clinic**, **decision-to-treat**, and **first definitive treatment** remain insufficiently visible, impairing the Trust’s ability to monitor compliance with the 31- and 62-day standards and to respond rapidly to operational pressures.

1.1 Problem Statement

The central problem addressed in this research is the absence of an integrated, operationally focused data infrastructure and decision-support interface to illuminate post-MDT workflow delays and guide timely interventions within an NHS oncology service. Despite the availability of relevant data, the lack of integration and harmonisation across MDT, clinic, and treatment

systems results in manual workarounds, inconsistent performance monitoring, and delayed insights. This gap diminishes the ability of operational teams to identify patients at risk of breaching national cancer standards, understand the factors driving variability in pathway intervals, and proactively manage capacity-constrained environments.

Moreover, existing reporting solutions tend to focus on aggregate compliance metrics after the fact, rather than providing real-time or near-real-time visibility of operational bottlenecks. Without a unified source of truth that aligns event definitions, timestamps, and clinical coding standards (NHS Digital, 2026), it becomes difficult to execute performance analysis or support day-to-day decision-making. Literature on Integrated Data Repositories (IDRs) similarly highlights that many institutions struggle with data harmonisation, terminological alignment, and the creation of analytics-ready architectures capable of supporting both research and operational use cases (Gagalova *et al.*, 2020).

This study therefore seeks to design, develop, and evaluate a **lightweight Integrated Data Repository (IDR)** and accompanying **decision-support dashboard** tailored specifically to the post-MDT oncology workflow. By integrating key pathway events, clinic capacity information, and national standard definitions, the artefact aims to surface bottlenecks, highlight breach risks, and support operational teams in prioritising actions that improve pathway performance.

1.2 Research Aim and Objectives

1.2.1 Research Aim

To design and evaluate an Integrated Data Repository and decision-support dashboard capable of improving operational visibility and proactive management of post-MDT oncology pathways.

1.2.2 Objectives

1. Identify and quantify post-MDT pathway delays.
2. Determine operational factors associated with delay.
3. Design an integrated data architecture.
4. Develop a user-centred dashboard.
5. Evaluate utility, usability and technical validity.
6. Derive transferable design principles

Research Question	DSR Activity	Primary Methods
RQ1: Where do delays occur between MDT outcome and treatment?	Problem identification	Descriptive statistics, interval analysis
RQ2: What operational factors predict extended delays?	Problem analysis	Regression and capacity analysis
RQ3: Does the artefact improve operational decision-making?	Evaluation	Task-based testing, time-to-answer, accuracy
RQ4: What design principles support post-MDT operational analytics?	Design knowledge	Literature synthesis and evaluation synthesis

Table 1.1: This mapping ensures methodological coherence between the research questions, artefact, and evaluation.

1.3 Research Approach

This study adopts Design Science Research (DSR) because the primary objective is not solely to explain an existing phenomenon but to create and evaluate an artefact intended to address an identified organisational problem. Whereas case-study research focuses on understanding contextual factors and Action Research emphasises organisational change through iterative intervention, DSR provides a structured framework for the design, construction and evaluation of innovative technological artefacts while simultaneously generating transferable knowledge (Hevner *et al.*, 2004; Peffers *et al.*, 2007).

The approach is particularly appropriate given the need to integrate heterogeneous operational data sources and translate them into actionable decision support for NHS oncology services. Similar DSR approaches have been successfully applied within health informatics, clinical decision support, and healthcare analytics domains where practical solutions and theoretical contribution are required concurrently.

The study therefore follows the six-stage Design Science Research Methodology (DSRM) proposed by Peffers *et al.* (2007), with detailed application described in Chapter 3.

1.4 Academic and Practical Significance

Academically, the research contributes to the limited body of DSR-grounded studies focusing on NHS operational analytics, offering:

- an IDR reference architecture for oncology operations;
- a set of design principles for NHS-appropriate decision-support dashboards;
- an evaluation protocol combining usability metrics, decision-quality testing, and technical verification.

Practically, the artefact provides operational teams with a tool that can:

- reduce time-to-insight;
- support rapid identification of bottlenecks and breach risks;
- align capacity planning with demand;
- improve pathway monitoring in line with the 28-, 31-, and 62-day standards.

Ultimately, the study aims to support faster, more reliable progression from MDT decision to treatment initiation, with the potential to support improved patient care and more effective management of oncology services.

DRAFT

Chapter 2

Literature Review

2.1 Introduction

The purpose of this literature review is to synthesise and critically evaluate existing research relevant to the design of an Integrated Data Repository (IDR) and operational analytics dashboard aimed at improving post-MDT oncology pathway efficiency. Timely cancer treatment has profound implications for survival, patient experience, resource utilisation, and system performance. Evidence from national datasets, observational studies, systematic reviews, and informatics research shows that delays between MDT decisions and treatment initiation remain persistent across the NHS. Furthermore, data fragmentation, inconsistent recording practices, and limited visibility of operational bottlenecks hinder the ability of clinicians, MDT coordinators, and service managers to proactively manage these delays.

The review is organised around five interrelated themes:

1. MDT operations and evolving pressures on multidisciplinary cancer decision making;
2. Operational delays across the cancer pathway, particularly diagnosis-to-treatment intervals (DTI);
3. Data fragmentation and the case for integrated data repositories;
4. Dashboards and performance-monitoring tools to support operational decision-making;
5. Emerging data trends and the role of ETL, interoperability, AI and multimodal data integration.

Collectively, this literature forms the conceptual foundation (kernel theory) for the artefact developed in this study. The following sections provide a detailed account of these themes, integrating high-quality evidence from the UK, international datasets, clinical oncology, health informatics, and organisational science.

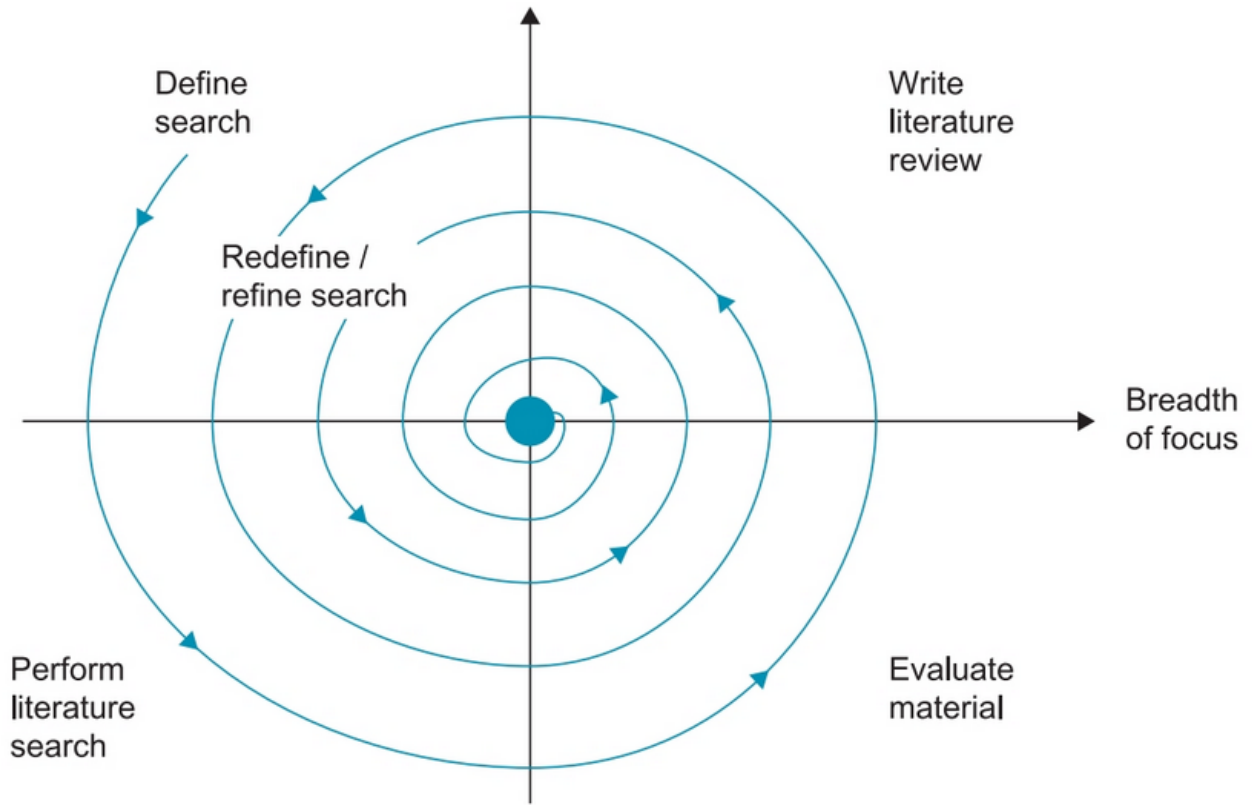


Figure 2.1: Iterative literature review process (Dawson, 2015)

2.1.1 Method

An iterative literature review was adopted, following the guidance outlined by Dawson (2015) (Figure 2.1). An initial search was conducted using Google Scholar and open-access repositories to identify foundational concepts and terminology related to Integrated Data Repositories, cancer pathways, MDT operations, dashboards, and Design Science Research. These preliminary results informed a more structured search strategy applied to ScienceDirect, IEEEXplore, and PubMed.

Searches were limited to peer-reviewed journals, systematic reviews, national audits, and authoritative policy documents published primarily between 2010 and 2026, with earlier seminal works retained where conceptually relevant (e.g. JCCO report (Ash *et al.*, 2004), the Calman-Hine report (Calman and Hine, 1995) and Design Science Research foundations). Studies were included if they addressed MDT processes, oncology pathway delays, health data integration, dashboards, usability standards, or operational analytics. Articles focused exclusively in molecular biology, single-modality clinical trials, or non-healthcare data systems were excluded.

Approximately 90 sources were reviewed during the process, with around 55 core studies retained for detailed analysis based on relevance, methodological quality, and applicability to NHS operational contexts. Backward citation tracking was used to identify influential seed papers and ensure conceptual completeness.

2.2 The NHS Cancer Pathway and MDT Framework

2.2.1 MDTs as the Backbone of UK Cancer Care

Since the publication of the Calman–Hine report (Calman and Hine, 1995) and subsequent interventions, MDTs have been embedded as the central governance structure for cancer decision making (Haward, 2006; Price *et al.*, 2014; Morris *et al.*, 2006, 2008). The MDT model mandates that every new cancer case in the UK is discussed by a multidisciplinary panel including oncologists, surgeons, radiologists, pathologists, specialist nurses, and allied professionals. The 2024 NHS Wales MDT Charter emphasises the crucial role MDTs play in standardising care, reducing unwarranted variation, and improving outcomes by bringing domain experts together in real time (NHS Wales, 2024). Prospective benefits include higher adherence to guidelines, improved communication, shared decision-making, and better coordination of multi-modal treatment plans.

However, the complexity of modern cancer care has increased dramatically alongside rising incidence, new treatment options, biomarker-based stratification, digital imaging volume, and genomic data availability. MDTs now contend with substantially larger case numbers, more complex diagnostic information, and resource pressures that stretch meeting capacity. Professional societies and national audits consistently highlight that MDT meetings are often over-subscribed, under-resourced, and administratively burdensome. These pressures compromise the quality of discussion and create operational bottlenecks that directly affect patient pathways.

2.2.2 Evidence of MDT Operational Pressures

Several studies provide evidence of the challenges affecting MDT functioning:

- Soukup *et al.* (2022) conducted a detailed observational analysis of 822 case discussions across three UK MDTs. Administrative and process issues occurred in 30% of cases, attendance problems in 16%, and equipment issues in 5%. These logistical challenges significantly reduced the quality of information and decision-making, demonstrating how operational inefficiencies detract from MDT effectiveness.
- Lim *et al.* (2025) analysed 1,488 UK Head and Neck Cancer patients across 50 MDTs and found extremely low compliance with national cancer pathway standards: 32.8% (FDS), 33.3% (31-day), and 34.6% (62-day). Median DTT-to-treatment was 42 days.
- Chakravarty *et al.* (2023) demonstrated that patients referred from secondary-care outpatient pathways experienced significantly longer delays to MDT discussion and treatment (mean 102 vs. 88 days). These patients were also more likely to receive palliative rather than curative treatment.

- Law *et al.* (2024) identified a decade of known MDT barriers including time pressures, communication challenges, inconsistent attendance, information deficits, and increasing clinical complexity.

While these studies consistently demonstrate operational strain within MDT processes, several limitations should be acknowledged. Much of the existing evidence is observational and focuses primarily on meeting effectiveness, communication quality, or decision-making processes rather than downstream operational consequences following MDT outcomes. For example, Soukup *et al.* (2022) identify logistical and administrative barriers that impair MDT performance, yet the study does not examine whether these issues translate into measurable delays in treatment initiation. Similarly, the scoping review proposed by Law *et al.* (2024) aggregates a broad range of organisational challenges but provides limited insight into which barriers exert the greatest influence on pathway performance.

Large-scale cohort studies such as Lim *et al.* (2025) quantify poor compliance with national standards, but their retrospective design offers little visibility of the specific workflow stages responsible for delay accumulation. Consequently, while the literature establishes that MDT processes operate under significant pressure, a gap remains in understanding how operational data can be used to identify, monitor, and mitigate delays arising after MDT decisions have been made. This distinction is important because improving MDT discussion quality alone may be insufficient if subsequent scheduling and treatment processes remain poorly coordinated.

2.2.3 National policy direction and waiting time standards

NHS England’s 2024/25 planning guidance emphasises compliance with the Faster Diagnosis Standard (FDS) and cancer treatment timeliness metrics. Persistent waiting-time challenges, data-quality requirements, and the need for integrated pathway oversight tools are highlighted (Parker, 2026). Radiotherapy services face particular strain: the Royal College of Radiologists reported over 2,300 breaches of the 31-day standard in October 2023 (RCR, 2024). Hidden waits—such as 20-week delays between surgery and radiotherapy—are not systematically captured, further illustrating limitations in operational visibility.

2.3 Operational Delays in Cancer Pathways

2.3.1 Diagnosis-to-Treatment Interval (DTI) as a Critical Metric

A robust body of literature demonstrates that prolonged DTI adversely affects survival across multiple cancer types.

Key findings include:

- Hanna *et al.* (2020): A systematic review of 34 studies (1.27M patients) found each four-week delay increased mortality by up to 6–8% across treatment modalities.

- Coca-Pelaz *et al.* (2018): Delays in Head and Neck Small Cell Carcinoma beyond four weeks contribute to tumour progression and reduced local control.
- Sharma *et al.* (2016): Delays greater than 30 days reduce overall survival (HR 1.12), with mortality increasing 2.2% per additional week.
- Villemure-Poliquin *et al.* (2025): Treatment within 30 days improves survival by 9%.
- Saleh *et al.* (2024): Cervical cancer survival drops dramatically after 30 days.

2.3.2 Hospital/System Factors Influencing Delay

Large-scale evidence shows that delays are driven not only by tumour characteristics but also by organisational inefficiencies:

- Yun *et al.* (2012): Delays over one month worsened outcomes, especially in low-volume hospitals.
- UK audits (e.g., Lim *et al.* (2025)) highlight delays in biopsies, imaging, scheduling, and treatment booking.

A further limitation of the current literature is the tendency to treat organisational performance as a static characteristic rather than a dynamic process. Factors such as hospital volume, resource availability, and service configuration are frequently associated with treatment delays; however, there is comparatively little exploration of how these risks evolve over time or how operational teams can intervene before delays become breaches. This reinforces the distinction between retrospective performance measurement and proactive pathway management.

2.3.3 MDT-specific drivers of post-decision delays

- Missing diagnostic information and administrative errors cause re-discussion delays (Soukup *et al.*, 2022).
- High caseloads reduce decision completeness.
- Secondary-care referrals introduce additional diagnostic delays (Chakravarty *et al.*, 2023).
- Capacity constraints (radiotherapy, theatres, oncology clinics) generate downstream bottlenecks.

Although the evidence linking treatment delay and mortality is compelling, caution should be exercised when applying these findings directly to operational improvement initiatives. Most studies evaluate aggregated diagnosis-to-treatment intervals and therefore provide limited insight into the relative contribution of individual pathway stages. While the findings clearly establish delay as clinically important, they do not identify where operational interventions are

likely to deliver the greatest benefit. From a service management perspective, this highlights the need for more granular operational analytics capable of decomposing pathway delays into actionable components.

2.3.4 Crititcal synthesis of delay literature

Collectively, the literature provides strong evidence that prolonged diagnosis-to-treatment intervals adversely affect patient outcomes across multiple cancer types. However, important limitations remain when attempting to translate this knowledge into operational practice. Most studies conceptualise delay as a single aggregated interval and consequently offer limited support for identifying specific workflow bottlenecks. Furthermore, while organisational variables such as hospital volume, diagnostic complexity, and service capacity are frequently identified as contributors to delay, relatively little attention has been given to the role of integrated operational data in supporting proactive intervention.

This omission is particularly significant within NHS oncology services, where national performance standards require action before breaches occur. Once a patient has experienced a clinically significant delay, retrospective performance reporting provides little opportunity to prevent harm. Accordingly, there is a need for analytical approaches capable of identifying emerging risks and supporting early operational intervention, rather than simply documenting historical performance.

2.4 Data Fragmentation and the Need for Integrated Data Repositories

2.4.1 Fragmentation Across NHS Systems

Oncology data spans EPRs, PAS, pathology, RIS/PACS, radiotherapy systems, chemotherapy e-prescribing, Somerset/Cosmic, and MDT spreadsheets.

Evidence includes:

- Operational policy from Royal Wolverhampton highlights inconsistent event definitions and PTL challenges (The Royal Wolverhampton NHS Trust, 2024).
- Varma *et al.* (2026): 92.3% agreement between CWT and hospital data only when treatment occurs within 100 days; discrepancies arise from definition mismatches and missing fields.

2.4.2 Clinical Data Warehouses (CDWs) and the Evolution Toward IDRs

Examples:

- ROOT CDW: real-time repository for 67,000+ cancer patients (Jung *et al.*, 2021).
- ONCO-FAIR: chemotherapy-specific FHIR extensions (Guilbert *et al.*, 2024).
- NHS-industry multimodal genomic-clinical data lake (Conway *et al.*, 2025).
- HN-XNAT: federated imaging and radiotherapy repository (Butterworth *et al.*, 2025).

While these examples demonstrate the technical feasibility of large-scale oncology data repositories, their transferability to routine NHS operational environments warrants careful consideration. Many reported implementations originate from digitally mature organisations with substantial informatics capability, dedicated technical teams, and highly structured data environments. Such conditions may not be representative of NHS Trusts where operational data are frequently distributed across legacy systems, locally developed databases, spreadsheets, and departmental applications.

Furthermore, the majority of published repositories prioritise research, population health, or precision medicine objectives. Although these use cases require extensive data integration, they often place less emphasis on the timeliness, operational responsiveness, and workflow visibility required for day-to-day pathway management. Consequently, a gap exists between the design of many clinical data repositories and the practical needs of operational cancer services.

2.4.3 Need for IDRs in NHS Operational Oncology

The absence of a robust IDR has implications beyond simple reporting inefficiency. Fragmented datasets create multiple versions of the truth, requiring operational staff to reconcile conflicting information from different systems before decisions can be made. This process introduces delays, increases administrative burden, and reduces confidence in performance measures.

Moreover, fragmented data environments encourage reactive management practices. Operational teams frequently identify pathway risks only after breaches have occurred because no single system provides a comprehensive view of patient progression across MDT, clinic, and treatment stages. As a result, opportunities for early intervention may be missed despite the underlying data already existing within organisational systems.

2.5 Dashboards, Operational Analytics, and Decision-Support Systems

2.5.1 The Role of Dashboards in Performance Management

Dashboards synthesise high-volume data for actionable insight. NHS guidance emphasises dashboards as essential for meeting FDS and treatment targets.

Buttigieg *et al.* (2017) identify:

- strategic dashboards,
- tactical dashboards,
- operational dashboards.

Key design principles include real-time refresh, drill-downs, user-centred design, and integration with data infrastructures.

Despite widespread adoption, dashboards are not universally associated with improved organisational performance. Several studies have reported that dashboards frequently become passive reporting tools rather than active decision-support systems. The mere presentation of information does not guarantee action, particularly when users are confronted with excessive metrics, poorly aligned visualisations, or data that lack clear operational relevance.

Consequently, dashboard effectiveness depends not only on technical implementation but also on the extent to which information is embedded within operational decision-making processes. This distinction is particularly relevant in cancer pathway management where users require actionable intelligence capable of informing scheduling, capacity allocation, and escalation decisions.

2.5.2 Clinical Decision-Support Dashboards and Multimodal Data

Examples:

- Multimodal CDSS supporting 170,000+ patients, using ETL + NLP + 140 data quality checks (Chang *et al.*, 2025).
- ROOT CDW: automated ingestion, structured/unstructured data integration (Jung *et al.*, 2021).

2.5.3 Data Quality and Informatics Challenges

Dashboards depend on strong data foundations:

- Coding variation limits dataset reliability (Varma *et al.*, 2026).
- Oncology workloads require custom FHIR extensions (Guilbert *et al.*, 2024).
- Multimodal data lakes require governance and standardisation (Conway *et al.*, 2025).

2.5.4 Reactive versus Proactive Operational Management

Traditional healthcare performance reporting has largely focused on retrospective measurement through key performance indicators and compliance monitoring. While such approaches provide valuable accountability mechanisms, they are inherently limited in their ability to prevent adverse outcomes because they identify problems only after they have occurred.

This limitation is particularly relevant in oncology services. Evidence demonstrates that delays in treatment initiation are associated with poorer patient outcomes (Hanna *et al.*, 2020; Sharma *et al.*, 2016). Consequently, a management approach based solely on retrospective review is problematic because a missed cancer target represents more than a performance failure; it may indicate a clinically significant delay experienced by a patient.

Continuous improvement frameworks such as Plan-Do-Check-Act (PDCA) remain widely used within healthcare quality improvement. However, PDCA assumes that outcomes can be observed, measured, and refined through successive iterations. In cancer pathway management, allowing delays to occur before evaluating performance may be ethically and operationally undesirable. Instead, there is a need for leading indicators capable of identifying emerging risks before breaches occur.

From an operational analytics perspective, this shifts the emphasis from historical reporting towards predictive and proactive decision support. Dashboards designed around breach-risk identification, remaining time-to-target, and capacity constraints may therefore provide greater practical value than systems focused exclusively on retrospective compliance measurement.

2.6 Usability, HCI, and ISO 9241-110

ISO 9241-110:2020 (International Organization for Standardization, 2020) provides seven principles:

1. Suitability for the task
2. Self-descriptiveness
3. Conformity with user expectations
4. Learnability
5. Controllability
6. Error tolerance
7. Suitability for individualisation

Fraunhofer usability research supports its relevance in healthcare (Hunkirchen).

2.7 AI-Enabled MDT Systems and Emerging Technologies

Examples include:

- EvoMDT: multi-agent MDT recommender reducing decision time by 30–40% (Liu *et al.*, 2026a).
- TrustedMDT: TNM staging and guideline checking integrated with Teams (Soltan *et al.*, 2025).
- AI-MDT for lung cancer: improved throughput and workload reduction (Liu *et al.*, 2026b).
- AI automation of transcription and summarisation (HealthOrbit, 2025).
- Multi-agent consensus MDT system achieving high inter-agent agreement (Han *et al.*, 2025).

While AI-enabled MDT systems report promising improvements in efficiency, consistency, and decision support, many studies remain limited by controlled implementation environments and highly structured datasets. Questions therefore remain regarding external validity and scalability within routine NHS settings characterised by fragmented information systems and variable data quality.

Importantly, the effectiveness of AI applications appears dependent upon the existence of integrated, standardised, and high-quality data architectures. Consequently, AI may be better viewed as a downstream capability rather than a standalone solution. Without addressing foundational challenges relating to interoperability, governance, and data integration, many of the anticipated benefits of AI-assisted MDT decision-making may remain difficult to realise in practice.

2.8 Synthesis of Gaps

The literature reveals a common pattern across otherwise diverse research domains. Studies examining MDT performance identify increasing operational pressure but provide limited visibility of downstream workflow delays. Research linking treatment delays to survival demonstrates the importance of timely treatment but offers limited guidance regarding how delays should be managed operationally. Similarly, data repository research demonstrates integration capability while dashboard studies emphasise information presentation; however, few studies combine these strands into a coherent operational solution.

The review therefore identifies three interconnected deficiencies. First, there is insufficient operational visibility across the post-MDT pathway. Second, existing approaches remain largely reactive, identifying problems after performance deterioration has already occurred. Third, there is limited evidence regarding how integrated operational data repositories and user-centred dashboards can be combined to support proactive cancer pathway management.

These gaps directly inform the design objectives of this study and provide the rationale for the Design Science Research approach described in Chapter 3.

2.9 Summary

This literature review synthesises evidence across MDT functioning, treatment delays, operational workflow challenges, data fragmentation, health informatics, dashboard design, usability standards, and emerging AI-driven MDT tools. MDTs remain vital but increasingly pressured. Delays between diagnosis, MDT decisions, and treatment initiation significantly worsen outcomes, yet Trusts lack the integrated systems required to manage these pathways proactively. Modern CDWs demonstrate feasible architectures, while dashboard and usability literature highlight design imperatives. AI-enabled MDT systems reveal future potential but highlight the foundational need for high-quality integrated data repositories.

The combined evidence exposes a clear research gap: NHS oncology services require a robust, user-centred, IDR-backed operational dashboard to improve MDT-to-treatment coordination.

Chapter 3

Methodology

3.1 Overview of Research Design

This research adopts a Design Science Research (DSR) methodology to address an operational problem in NHS oncology services: the lack of integrated, actionable visibility of delays between MDT outcomes and treatment initiation. DSR is particularly suitable for this study because it explicitly focuses on the design, construction, and evaluation of an artefact intended to improve a real-world organisational context, while generating transferable design knowledge (Hevner *et al.*, 2004).

The primary artefact developed in this study is a lightweight integrated data repository (IDR) and decision-support dashboard, designed to consolidate post-MDT pathway data and support operational decision-making. In addition to the artefact itself, the research produces a set of evaluated design principles for oncology operational analytics, constituting a contribution to knowledge beyond the local implementation.

3.1.1 Justification for Design Science Research

Several methodological approaches were considered. A case study methodology would have enabled detailed examination of post-MDT operational processes but would not have provided a structured framework for the design and evaluation of an intervention. Action Research was also considered because of its focus on organisational change and practitioner involvement. However, Action Research typically requires repeated cycles of intervention and measurement within live operational environments. In cancer pathway management, allowing breaches to occur in order to evaluate an intervention would be ethically and operationally undesirable given the known relationship between treatment delays and patient outcomes.

Design Science Research was therefore selected because it provides a rigorous framework for designing, developing, and evaluating an artefact intended to solve a real-world organisational problem while simultaneously generating transferable knowledge. DSR has been widely applied within healthcare informatics, clinical decision support, and data integration research where new technological artefacts are required to address operational challenges (Hevner *et al.*, 2004;

Peffers *et al.*, 2007).

Unlike traditional continuous-improvement approaches such as Plan-Do-Check-Act (PDCA), cancer pathway management cannot rely solely on post-hoc measurement of outcomes. A breach of the 31-day or 62-day standard represents a clinically meaningful delay that may adversely affect patient outcomes. Consequently, operational management must emphasise proactive identification and mitigation of risk rather than retrospective evaluation alone. The design objective of the proposed dashboard is therefore not simply to identify breaches after they occur, but to provide leading indicators that enable intervention before delays become clinically significant.

Case study research was rejected because the objective of this study extends beyond understanding operational delays to designing and evaluating a technological intervention intended to address them. Similarly, Action Research was considered unsuitable because it emphasises organisational change through repeated intervention cycles, whereas the primary contribution of this study is the creation and evaluation of an information systems artefact.

3.2 Design Science Research Framework

The study follows the Design Science Research Methodology (DSRM) proposed by Peffers *et al.* (2007), which structures DSR into six interrelated activities. These activities are applied iteratively rather than strictly linearly.

3.2.1 Problem Identification and Motivation

The problem identification phase was undertaken using a combination of retrospective pathway analysis, operational observations, and literature synthesis. Local cancer waiting time reports and patient tracking data were reviewed to identify recurring delays between MDT outcomes, first oncology consultation, decision-to-treat, and treatment initiation. Findings were triangulated with evidence from the literature review, which demonstrated similar challenges across NHS organisations. This process established the need for an integrated operational analytics solution capable of identifying bottlenecks before breaches occur.

3.2.2 Objectives of a Solution

Solution objectives were derived through a requirements elicitation process combining literature-derived design requirements, operational needs identified through pathway review, and information requirements associated with national cancer waiting time standards. Requirements were grouped into data integration, operational monitoring, decision support, and usability domains.

3.2.3 Design and Development

The design and development activity involved two tightly coupled components:

1. Integrated Data Repository (IDR)
2. Decision-Support Dashboard

The IDR adopts a general-model architecture consisting of a staging layer for ETL and harmonisation, a central warehouse, and an application layer for analytics and visualisation, consistent with IDR literature for single-institution deployments (Gagalova *et al.*, 2020).

The dashboard is designed to support operational tasks rather than retrospective reporting, aligning with the role of operational dashboards in healthcare performance management (Buttigieg *et al.*, 2017).

Both components were developed iteratively, guided by explicit design principles derived from prior research and refined during prototyping.

3.2.4 Demonstration

The artefact is demonstrated using:

- retrospective Trust data covering MDT decisions, clinic appointments, and treatment initiation;
- realistic operational scenarios reflecting common management tasks (e.g., identifying upcoming breach risks, assessing clinic capacity).

Demonstration focuses on illustrating how the artefact supports decision-making that was previously manual, fragmented, or retrospective.

3.2.5 Evaluation

Evaluation is multi-method and aligned with Hevner *et al.* (2004)'s guidance that DSR artefacts must be assessed for utility, quality, and efficacy. Evaluation methods are detailed in Section 3.4.

3.2.6 Communication

The final activity consists of communicating:

- the artefact,
- its measured effects, and
- the derived design principles,

through this dissertation, targeting both academic and practitioner audiences.

3.3 Design Principles

Design principles represent the design knowledge contribution of this study. They are formulated using a standard DSR structure: *context*, *intervention*, and *rationale*. These initial principles are refined following evaluation.

DP1: Surface Breach Risks as a Leading Indicator

If operational teams must prioritise actions in time-constrained cancer pathways, then dashboards should surface leading indicators of breach risk (e.g. days remaining to breach thresholds), because prospective visibility enables earlier and more effective intervention than retrospective compliance metrics.

DP2: Pair Risk Information with Actionable Capacity Data

If users are expected to act on identified risks, then risk indicators should be presented alongside relevant capacity and demand information, because risk awareness without visible operational levers does not support decision-making.

DP3: Use Canonical, Standard-Aligned Pathway Events

If pathway metrics are to be trusted and comparable, then all intervals should be derived from canonical event definitions aligned to NCWTMDS rules, because inconsistent definitions undermine credibility and evaluation validity.

DP4: Match Data Refresh Cadence to the Action Horizon

If decisions depend on short operational windows, then data refresh frequency should be aligned to the decision horizon (e.g. daily or near-real-time where feasible), because stale information degrades decision quality.

DP5: Preserve Data Provenance and Auditability

If analytics are deployed in regulated healthcare environments, then all derived metrics must maintain transparent provenance to source systems, because trust, governance, and reconciliation are essential for adoption.

DP6: Align Interaction Design with Real Operational Tasks

If dashboards are intended to support complex operational work, then interaction design should reflect users' real task sequences, because task-aligned interfaces reduce cognitive load and improve decision quality (International Organization for Standardization, 2020).

These principles directly inform both the IDR architecture and the dashboard design, and serve as evaluation criteria in Section 3.4.

3.4 Evaluation Strategy

Evaluation combines formative and summative methods across four dimensions.

3.4.1 Usability Evaluation

- System Usability Scale (SUS)
- Heuristic review grounded in ISO 9241-110
- Think-aloud walkthroughs

3.4.2 Utility and Effectiveness

- Task-based experiment with operational users
- Primary measures:
 - time-to-answer key operational questions,
 - accuracy of breach-risk identification,
 - decision quality in constrained scheduling scenarios

3.4.3 Technical Validity

- Reconciliation of dashboard metrics with source SQL queries
- Validation of interval derivations against NCWTMDS rules

3.4.4 Design Knowledge Evaluation

- Synthesis of user feedback and performance outcomes
- Refinement and confirmation of final design principles

3.5 Ethical and Governance Considerations

The study operates within UK GDPR and NHS data governance requirements. All data are de-identified or pseudonymised, with access controls and separation between operational and research use. The project follows university ethics procedures and local Trust information governance guidance.

3.5.1 Research Limitations

- **Single-site study:** The artefact will be developed and evaluated within a single NHS Trust, which may limit generalisability.
- **Data quality and availability:** Findings depend on the completeness and accuracy of source systems.
- **Retrospective data:** Restrospective data may not fully represent future operational conditions.
- **Usability sample size:** Evaluation participants are likely to represent a relatively small operational user population.
- **Artificial evaluation environment:** Demonstration scenarios may not fully replicate live decision-making environments.
- **DSR limitation:** Evaluation assesses utility and usability rather than directly measuring improvements in patient outcomes.

3.6 Summary

This chapter has outlined a rigorous, design-science-based methodology that supports both artefact creation and theory-driven knowledge contribution. By embedding explicit design principles within the DSR framework and evaluating the artefact using multiple complementary methods, the study ensures methodological robustness, practical relevance, and academic contribution.

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List of Acronyms

AI Artificial Intelligence

CDSS Clinical Decision Support System

CDW Clinical Data Warehouse

CWT Cancer Wait Time

DSR Design Science Research

DSRM Design Science Research Methodology

DTI Diagnosis to Treatment Interval

DTT Decision to Treat to Treatment

ETL Extract-Transform-Load

EPR Electronic Patient Record

FDS Faster Diagnosis Standard

FHIR Fast Healthcare Interoperability Resources

HCI Human Computer Interaction

HNC Head and Neck Cancer

HNSCC Head and Neck Squamous Cell Carcinoma

HR Hazard Ratio

IDR Integrated Data Repository

MDT Multidisciplinary Team

NCWTMDS National Cancer Waiting Times Monitoring Data Set

NHS National Health Service

NLP Natural Language Processing

PACS Picture Archiving and Communication System

PAS Patient Administration System

PTL Patient Tracking List

RCR Royal College of Radiologists

RIS (Radiology Information System)

RTT Referral To Treatment

DRAFT